Topics in Algebraic Structures

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Abstract

Contents

Part I

Graded Extensions of Familiar Constructs

1 Tensor, Symmetric and Exterior Algebras over Graded Modules

1.1 The Free Graded Algebra

Definition 1.1.1 (The Free Graded Algebra)
Let R be a commutative ring. Let M be a graded R-module. Define the free graded algebra of M to be

$$T^{\text{gr}}(M) = \bigoplus_{i=0}^{\infty} M^{\otimes i}$$

together with grading given by $\deg(v_1 \otimes \cdots \otimes v_n) = \deg(v_1) + \cdots + \deg(v_n)$.

Proposition 1.1.2 (Universal Property)

1.2 Symmetric Algebras over Graded Modules

We have seen the tensor, symmetric and exterior algebra of an R-module. When the underlying R-module is graded, these algebra become bigraded and the theory differs slightly.

Definition 1.2.1 (The Graded Braid Map)
Let R be a commutative graded ring. Let M be a graded R-module. Define the graded braid map $\tau : M \otimes_R M \to M \otimes_R M$ by

$$\tau(x \otimes y) = (-1)^{ij} y \otimes x$$

for $x \in M_i$ and $y \in M_j$.

Definition 1.2.2 (The Symmetric Algebra of a Graded Module)
Let R be a commutative graded ring. Let M be a graded R-module. Define the symmetric algebra of M to be the quotient

$$S^{\text{gr}}(M) = \frac{T^{\text{gr}}(M)}{\langle x \otimes y - \tau(x \otimes y) \mid x, y \in M \rangle}$$

Proposition 1.2.3
Let R be a commutative graded ring. Let M be a graded R-module. Then $S^{\text{gr}}(M)$ is a graded commutative R-module.

Proposition 1.2.4
Let R be a commutative graded ring. Let M be a graded R-module. Then there is a graded R-isomorphism

$$S^{\text{gr}}(M) \cong S(M_{\text{even}}) \otimes_R \Lambda(M_{\text{odd}})$$

1.3 Exterior Algebras over Graded Modules

Definition 1.3.1 (The Exterior Algebra of a Graded Module)
Let R be a commutative graded ring. Let M be a graded R-module. Define the exterior algebra of M to be the quotient

$$\Lambda^{\text{gr}}(M) = \frac{T^{\text{gr}}(M)}{\langle x \otimes y + \tau(x \otimes y), x \otimes x \mid x, y \in M \rangle}$$

Proposition 1.3.2
Let R be a commutative graded ring. Let M be a graded R-module. Then there is a graded R-isomorphism

$$\Lambda^{\text{gr}}(M) \cong \Lambda(M_{\text{even}}) \otimes_R S(M_{\text{odd}})$$

2 Graded Lie Algebras

Definition 2.0.1 (Graded Lie Algebras)
Let R be a commutative ring. Let $M = \bigoplus_{i=0}^{\infty} M_i$ be a graded R-module. We say that M is a graded Lie algebra if there exists an R-module homomorphism $[-,-] : M_i \times_j \to M_{i+j}$ such that the following are true.

- $[x,y] = (-1)ij + 1[y,x]$ for all $x \in M_i$ and $y \in M_j$.
- Jacobi Identity: $[x,[y,z]] = [[x,y],z] + (-1)^{ij}[y,[x,z]]$.
- $[x,[x,x]] = 0$.

Part II

Categorical Algebra

Want: group objects, ring objects, algebra objects, commutative algebra objects, monoid objects, module objects, group objects acting on objects. morphisms between these objects.

3 The Theory of Semigroups

3.1 Basic Properties of Semigroups

Definition 3.1.1 (Semigroups)
A semigroup is a pair $(S, *)$ where S is a set and $* : S \times S \to S$ is a binary operation on S such that
$$(a * b) * c = a * (b * c)$$
for all $a, b, c \in S$.

Definition 3.1.2 (Semigroup Homomorphisms)
Let S, T be semigroups. Let $\phi : S \to T$ be a function. We say that ϕ is a semigroup homomorphism if
$$\phi(ab) = \phi(a)\phi(b)$$
for all $a, b \in S$.

Definition 3.1.3 (Commutative Semigroups)
Let S be a semigroup. We say that S is commutative if
$$a * b = b * a$$
for all $a, b \in S$.

Definition 3.1.4 (Cancellative Semigroups)
Let S be a semigroup. We say that S is cancellative if the following are true.
- $ab = ac$ implies $b = c$ for all $a, b, c \in S$.
- $ba = ca$ implies $b = c$ for all $a, b, c \in S$.

Definition 3.1.5 (Finitely Generated Semigroups)
Let S be a semigroup. We say that S is finitely generated if there exists $a_1, \dots, a_n \in S$ such that
$$a = a_1^{k_1} \cdots a_n^{k_n}$$
for all $a \in M$ and some $k_1, \dots, k_n \in \mathbb{N}$.

3.2 The Semigroup Ring

Definition 3.2.1 (The Semigroup Ring)
Let S be a semigroup. Let R be a ring. Define the semigroup ring of S over R to be the left R-module
$$R[S] = \bigoplus_{s \in S} R \cdot t^s$$
together with multiplication defined on basis by $t^{s_1} \cdot t^{s_2} = t^{s_1 + s_2}$

Lemma 3.2.2
Let S be a semigroup. Let R be a ring. Then the following are true.
- $R[S]$ is a ring.
- If S is commutative, then $R[S]$ is a commutative ring.
- If R is commutative, then $R[S]$ is an R-algebra.
- If R and S are commutative, then $R[S]$ is a commutative R-algebra.

4 The Theory of Monoids

4.1 Basic Properties of Monoids

Definition 4.1.1 (Monoids)
A monoid is a pair $(M, *)$ where M is a set and $* : M \times M \to M$ is a binary operation on M satisfying the following axioms:
- Associativity: $(a * b) * c = a * (b * c)$ for all $a, b, c \in M$.
- Identity: There exists $1_M \in M$ such that $1_M * a = a = a * 1_M$ for all $a \in M$.

Definition 4.1.2 (Monoid Homomorphisms)
Let M, N be monoids. Let $\phi : M \to N$ be a function. We say that ϕ is a monoid homomorphism if the following are true.
- $\phi(m_1 m_2) = \phi(m_1)\phi(m_2)$ for all $m_1, m_2 \in M$.
- $\phi(1_M) = 1_N$.

Definition 4.1.3 (The Category of Monoids)

Definition 4.1.4 (The Category of Commutative Monoids)

4.2 The Grothendieck Completion of a Group

Definition 4.2.1 (Grothendieck Completion)
Let M be a commutative monoid. Define the Grothendieck completion of M to be the set

$$G(M) = \frac{M \times M}{\sim}$$

where $(m_1, m_2) \sim (n_1, n_2)$ if $m_1 + n_2 + k = m_2 + n_1 + k$ together with the binary operation $* : G(M) \times G(M) \to G(M)$ defined by

$$(m_1, m_2) * (n_1, n_2) = (m_1 + n_1, m_2 + n_2)$$

Lemma 4.2.2
Let M be a commutative monoid. Then $G(M)$ is an abelian group.

Definition 4.2.3 (The Canonical Homomorphism)
Let M be a commutative monoid. Define the canonical homomorphism to the Grothendieck completion to be the function $i : M \to G(M)$ given by $i(m) = (m, 0)$.

Proposition 4.2.4 (Universal Property of the Grothendieck Completion)
Let M be a commutative monoid. Then $G(M)$ and the canonical map $i : M \to G(M)$ satisfies the following universal property: For any abelian group G and any monoid homomorphism $\phi : M \to G$, there exists a unique group homomorphism $\psi : G(M) \to G$ such that the following diagram commutes:

$$\begin{tikzcd} M \arrow[r, "i"] \arrow[dr, "\phi"'] & G(M) \arrow[d, dashed, "\exists!\psi"] \\ & G \end{tikzcd}$$

Moreover, $G(M)$ is the unique (up to unique isomorphism) abelian group satisfying this property.

Lemma 4.2.5
Let M be a cancellative commutative monoid. Then the canonical morphism $i : M \to G(M)$ is injective.

Proposition 4.2.6

There is an adjunction $G : \mathbf{CMon} \overset{\dashv}{\rightleftarrows} \mathbf{Ab} : U$ with the forgetful functor. Explicitly, there is a natural isomorphism

$$\text{Hom}_{\mathbf{Ab}}(G(M), A) \cong \text{Hom}_{\mathbf{CMon}}(M, A)$$

4.3 Monoid Objects

Definition 4.3.1 (Monoid Objects)
Let $(\mathcal{C}, \otimes, 1)$ be a monoidal category. A monoid object in $\mathcal{C}$ is an object $M \in \mathcal{C}$ together with the following data:

- A morphism $\mu : M \otimes M \to M$ called multiplication.
- A morphism $\eta : I \to M$ called the unit.

such that the following diagrams commute:

$$\begin{tikzcd} M \otimes M \otimes M \arrow[r, "\mu\otimes\text{id}_M"] \arrow[d, "\text{id}_M\otimes\mu"'] & M \otimes M \arrow[d, "\mu"] \\ M \otimes M \arrow[r, "\mu"] & M \end{tikzcd} \qquad \begin{tikzcd} M \arrow[r, "(\text{id}_M{,}\eta)"] \arrow[d, "(\eta{,}\text{id}_M)"'] \arrow[dr, "\text{id}_M"] & M \otimes M \arrow[d, "\mu"] \\ M \otimes M \arrow[r, "\mu"'] & M \end{tikzcd}$$

Proposition 4.3.2
Let $(\mathcal{C}, \otimes, I)$ be a monoidal category. Let $X \in \mathcal{C}$ be an object. Then X is a monoid object if and only if the Yoneda embedding $h_X : \mathcal{C} \to \mathbf{Set}$ given by $C \mapsto \text{Hom}_{\mathcal{C}}(C, X)$ factors through the forgetful functor $\mathbf{Mon} \to \mathbf{Set}$.

Definition 4.3.3 (Morphisms of Monoid Objects)
Let $(\mathcal{C}, \otimes, I)$ be a monoidal category. Let $M, N \in \mathcal{C}$ be monoid objects. Let $\varphi : M \to N$ be a morphsim in $\mathcal{C}$. We say that φ is a morphism of monoid objects if

$$\varphi \circ \mu_M = \mu_N \circ (\varphi \otimes \varphi)$$

Lemma 4.3.4
Let $(\mathcal{C}, \otimes, I)$ be a monoidal category. Let $M, N \in \mathcal{C}$ be monoid objects. Let $\varphi : M \to N$ be a morphsim of monoid objects. Then we have

$$\varphi \circ \eta_M = \eta_N \circ \varphi$$

Proposition 4.3.5
Let $(\mathcal{C}, \otimes, I)$ be a monoidal category. Let $M, N \in \mathcal{C}$ be monoid objects. Let $\varphi : G \to H$ be a morphsim in $\mathcal{C}$. Then φ is a morphism of monoid objects if and only if the Yoneda embedding

$$h_\varphi(C) : \text{Hom}_{\mathcal{C}}(C, G) \to \text{Hom}_{\mathcal{C}}(C, H)$$

is a monoid homomorphism for all $C \in \mathcal{C}$.

Definition 4.3.6 (The Category of Monoid Objects)
Let $(\mathcal{C}, \otimes, I)$ be a monoidal category. Define the category of monoid objects $\mathbf{Mon}(\mathcal{C})$ by the following data.

- The objcets are the monoid objects of $\mathcal{C}$.
- The morphisms are the morphisms of monoid objects.
- Composition is given by composition in $\mathcal{C}$.

Example 4.3.7
The following are true.

- $\mathbf{Mon}(\mathbf{Set}) = \mathbf{Mon}$.
- $\mathbf{Mon}(\mathbf{Mon}) = \mathbf{CMon}$.

- $\mathbf{Mon}(\mathbf{Ab}, \otimes_{\mathbb{Z}}, \mathbb{Z}) = \mathbf{Ring}$.
- Let R be a commutative ring. Then $\mathbf{Mon}({}_R\mathbf{Mod}, \otimes_R, R) = \mathbf{Alg}_R$.
- Let R be a commutative ring. Then $\mathbf{Mon}({}_R\mathbf{GrMod}, \otimes_R, R) = \mathbf{GrAlg}_R$.

4.4 Commutative Monoid Objects

Definition 4.4.1 (Commutative Monoid Objects)
Let $(\mathcal{C}, \otimes, I, \tau)$ be a symmetric monoidal category. Let $M \in \mathcal{C}$ be a monoid object. We say that M is a commutative monoid object if

$$\mu \circ \tau_{M,M} = \mu$$

Definition 4.4.2 (The Category of Commutative Monoid Objects)
Let $(\mathcal{C}, \otimes, I, \tau)$ be a symmetric monoidal category. Define category of commutative monoid objects to be the full subcategory

$$\mathbf{CMon}(\mathcal{C}) \subseteq \mathbf{Mon}(\mathcal{C})$$

consisting of commutative monoid objects.

Example 4.4.3
The following are true.
- $\mathbf{CMon}(\mathbf{Set}) = \mathbf{CMon}$.
- $\mathbf{CMon}(\mathbf{Ab}, \otimes_{\mathbb{Z}}, \mathbb{Z}) = \mathbf{CRing}$.
- Let R be a commutative ring. Then $\mathbf{CMon}({}_R\mathbf{Mod}, \otimes_R, R) = \mathbf{CAlg}_R$.

4.5 Action Objects

Definition 4.5.1 ((Left) Action Objects)
Let $(\mathcal{C}, \otimes, I)$ be a monoidal category. Let $M \in \mathcal{C}$ be a monoid object. Let $X \in \mathcal{C}$. An action of M on X is a morphism $\alpha : M \times X \to X$ such that the following diagrams are commutative:

$$\begin{tikzcd} M \otimes M \otimes X \arrow[r, "\mathrm{id}_M \otimes \alpha"] \arrow[d, "\mu \otimes \mathrm{id}_X"'] & M \otimes X \arrow[d, "\alpha"] \\ M \otimes X \arrow[r, "\alpha"'] & X \end{tikzcd} \qquad \begin{tikzcd} 1 \otimes X \arrow[r, "\eta \otimes \mathrm{id}_X"] \arrow[d, "\cong"'] & M \otimes X \arrow[d, "\alpha"] \\ X \arrow[r, "\mathrm{id}_X"'] & X \end{tikzcd}$$

In this case we call X an M-action object.

Yoneda embedding counterpart.

Definition 4.5.2 (Morphisms of Action Objects)
Let $(\mathcal{C}, \otimes, I)$ be a monoidal category. Let $M \in \mathcal{C}$ be a monoid object. Let $X, Y \in \mathcal{C}$ be M-action objects. Let $f : X \to Y$ be a morphism in $\mathcal{C}$. We say that f is a morphism of M-action objects if

$$f \circ \alpha_X = \alpha_Y \circ (\mathrm{id}_M \otimes f)$$

Definition 4.5.3 (The Category of Action Objects)
Let $(\mathcal{C}, \otimes, I)$ be a monoidal category. Let $M \in \mathcal{C}$ be a monoid object. Define the category of left M-action objects ${}_M\mathbf{Act}(\mathcal{C})$ by the following data.
- The objects are the M-action objects.
- The morphisms are the morphism of M-action objects.
- Composition is given by composition in $\mathcal{C}$.

Example 4.5.4
The following are true.
- Action objects in $\mathbf{Set}$ are M-sets.
- Action objects in $\mathbf{Ab}$ are modules over a ring.

- Action objects in ${}_R\mathbf{Mod}$ are R-algebras.

Proposition 4.5.5
Let $(\mathcal{C}, \otimes, I)$ be a closed monoidal category. Let $M \in \mathcal{C}$ be a monoid object. Let $X \in \mathcal{C}$. Let $\alpha : M \times X \to X$ be a morphism. Then α is a monoid action if and only if the curry of α given by $M \to \text{HOM}(X, X)$ is a monoid homomorphism.

Define the opposite of a monoid (it is also a monoid).

Prove that ${}_M\mathbf{Act}(\mathcal{C}) = \mathbf{Act}(\mathcal{C})_M$.

Definition 4.5.6 ((Right) Action Objects)

Definition 4.5.7 (Morphims of Right Action Objects)

Definition 4.5.8 (The Category of Right Action Objects)

Proposition 4.5.9
Let $(\mathcal{C}, \otimes, I)$ be a monoidal category. Let $M \in \mathcal{C}$ be a monoid object. Then there is a natural equivalence of categories
$${}_M\mathbf{Mon}(\mathcal{C}) \simeq \mathbf{Mon}(\mathcal{C})_M$$

5 Group Objects

5.1 Basic Properties

Definition 5.1.1 (Group Objects)
Let $(\mathcal{C}, \otimes, 1)$ be a monoidal category. A group object in $\mathcal{C}$ is an monoid object $G \in \mathcal{C}$ together a morphism $\text{inv} : G \to G$ called the inverse such that the following diagram commutes:

$$\begin{tikzcd} G \arrow[r, "(\text{id}_G, \text{inv})"] \arrow[d, "(\text{inv}, \text{id}_G)"'] \arrow[dr, "\eta\circ p"] & G \otimes G \arrow[d, "\mu"] \\ G \otimes G \arrow[r, "\mu"'] & G \end{tikzcd}$$

where $p : G \to I$ is the unique terminal morphism.

Proposition 5.1.2
Let $(\mathcal{C}, \otimes, I)$ be a monoidal category. Let (M, μ, η) be a monoid object in $\mathcal{C}$. Then the following are equivalent.

- M is a group object.
- The following diagram is a pullback square:

$$\begin{tikzcd} M \otimes M \otimes G \arrow[r, "\mu\otimes\text{id}_M"] \arrow[d, "\text{id}_M\otimes\mu"'] & M \otimes M \arrow[d, "\mu"] \\ M \otimes M \arrow[r, "\mu"] & M \end{tikzcd}$$

- The Yoneda embedding $h_M : \mathcal{C} \to \textbf{Set}$ given by $C \mapsto \text{Hom}_{\mathcal{C}}(C, M)$ factors through the forgetful functor $\textbf{Grp} \to \textbf{Set}$.

Definition 5.1.3 (Morphisms of Group Objects)
Let $(\mathcal{C}, \otimes, I)$ be a monoidal category. Let $G, H \in \mathcal{C}$ be group objects. Let $\varphi : G \to H$ be a morphsim in $\mathcal{C}$. We say that φ is a morphism of group objects if it is a morphism of monoid objects.

Lemma 5.1.4
Let $(\mathcal{C}, \otimes, I)$ be a monoidal category. Let $G, H \in \mathcal{C}$ be group objects. Let $\varphi : G \to H$ be a morphsim of group objects. Then the following are true.

- $\varphi \circ \eta_G = \eta_H \circ \varphi$.
- $\varphi \circ \text{inv}_G = \text{inv}_H \circ \varphi$

Proposition 5.1.5
Let $(\mathcal{C}, \otimes, I)$ be a monoidal category. Let $G, H \in \mathcal{C}$ be group objects. Let $\varphi : G \to H$ be a morphsim in $\mathcal{C}$. Then φ is a morphism of group objects if and only if the Yoneda embedding

$$h_\varphi(C) : \text{Hom}_{\mathcal{C}}(C, G) \to \text{Hom}_{\mathcal{C}}(C, H)$$

is a group homomorphism for all $C \in \mathcal{C}$.

Definition 5.1.6 (The Category of Group Objects)
Let $(\mathcal{C}, \otimes, I)$ be a monoidal category. Define the category of group objects $\textbf{Grp}(\mathcal{C})$ by the following data.

- The objcets are the group objects of $\mathcal{C}$.
- The morphisms are the morphisms of group objects.
- Composition is given by composition in $\mathcal{C}$.

Example 5.1.7
A group object in $\textbf{Set}$ is a group.

two compatible group structure imply they coincide and abelian special case: group object in $\textbf{Grp}$ is

Ab.

5.2 Abelian Group Objects

5.3 Action of a Group Object

Definition 5.3.1 (Action of a Group Object)
Let $(\mathcal{C}, \otimes, I)$ be a monoidal category. Let $G \in \mathcal{C}$ be a group object. Let $X \in \mathcal{C}$. An action of G on X is a monoid action $\alpha : G \times X \to X$. In this case we call X a G-action object.

Again there is a Yoneda counterpart.

Also group action objects are exactly the same as monoid action objects, nothing changed.

Definition 5.3.2 (Morphisms of Action Objects)
Let $(\mathcal{C}, \otimes, I)$ be a monoidal category. Let $G \in \mathcal{C}$ be a group object. Let $X, Y \in \mathcal{C}$ be G-action objects. Let $f : X \to Y$ be a morphism in $\mathcal{C}$. We say that f is a morphism of G-action objects if it is a morphism of G-module objects.

The category is now denoted by $\mathbf{Act}_{\mathcal{C}}(G)$.

Under assumptions (eg finite limits exists), we can define fix points, orbits, stabilizers etc.

6 Ring, Module and Algebra Objects

6.1 Ring Objects

Definition 6.1.1 (The Category of Ring Objects)
Let $(\mathcal{C}, \otimes, I)$ be a monoidal category. Define the category of ring objects by

$$\mathbf{Ring}(\mathcal{C}) = \mathbf{Mon}(\mathbf{Ab}(\mathcal{C}))$$

Note: group objects of group objects = monoid objects of group objects = group objects of monoid objects = abelian group objects. So ring objects are constructed from monoids and groups. Similarly, we will see this for algebra objects.

Definition 6.1.2 (The Category of Commutative Ring Objects)
Let $(\mathcal{C}, \otimes, I)$ be a symmetric monoidal category. Define the category of commutative ring objects by

$$\mathbf{CRing}(\mathcal{C}) = \mathbf{CMon}(\mathbf{Ab}(\mathcal{C}))$$

6.2 Module Objects

Definition 6.2.1 ((Left) Module Objects)
Let $(\mathcal{C}, \otimes, I)$ be a monoidal category. Let $R \in \mathbf{Ring}(\mathcal{C})$ be a ring object. Define the category of left module objects by

$${}_R\mathbf{Mod}(\mathcal{C}) = {}_R\mathbf{Act}(\mathbf{Ab}(\mathcal{C}))$$

6.3 Algebra Objects

Definition 6.3.1 (The Category of Algebra Objects)
Let $(\mathcal{C}, \otimes, I)$ be a monoidal category. Let $R \in \mathbf{CRing}(\mathcal{C})$ be a commutative ring object. Define the category of algebra objects over R by

$$\mathbf{Alg}_R(\mathcal{C}) = \mathbf{Mon}({}_R\mathbf{Mod}(\mathcal{C}))$$

Proposition 6.3.2
Let $(\mathcal{C}, \otimes, I)$ be a monoidal category. Let $R \in \mathbf{CRing}(\mathcal{C})$ be a commutative ring object. Then there is an equivalence of categories

$$\mathbf{Alg}_R(\mathcal{C}) \simeq R/\mathbf{Mon}(\mathcal{C})$$

Definition 6.3.3 (The Category of Algebra Objects)
Let $(\mathcal{C}, \otimes, I)$ be a symmetric monoidal category. Let $R \in \mathbf{CRing}(\mathcal{C})$ be a commutative ring object. Define the category of algebra objects over R by

$$\mathbf{CAlg}_R(\mathcal{C}) = \mathbf{CMon}({}_R\mathbf{Mod}(\mathcal{C}))$$

Proposition 6.3.4
Let $(\mathcal{C}, \otimes, I)$ be a symmetric monoidal category. Let $R \in \mathbf{CRing}(\mathcal{C})$ be a commutative ring object. Then there is an equivalence of categories

$$\mathbf{CAlg}_R(\mathcal{C}) \simeq R/\mathbf{CMon}(\mathcal{C})$$

6.4 Graded Objects

6.5 Introduction to Operad Theory

Part III
Duals of Categorical Algebra

Want: comodule objects, coalgebra objects.

7 Comodules and Coalgebras

7.1 Coalgebras

Definition 7.1.1 (Coalgebras)
Let R be a commutative ring. A coalgebra over R is an R-module C together with the following data.
- An R-module homomorphism $\Delta : C \to C \otimes_R C$ called comultiplication.
- An R-module homomorphism $\varepsilon : C \to R$ called counit.

such that the following diagrams are commutative:

$$\begin{tikzcd} C \arrow[r, "\Delta"] \arrow[d, "\Delta"'] & C \otimes_R C \arrow[d, "\mathrm{id}_C \otimes \Delta"] \\ C \otimes_R C \arrow[r, "\Delta \otimes \mathrm{id}_C"'] & C \otimes_R C \otimes_R C \end{tikzcd} \qquad \begin{tikzcd} C \arrow[r, "\Delta"] \arrow[d, "\Delta"'] \arrow[dr, "\mathrm{id}_C"] & C \otimes_R C \arrow[d, "\varepsilon \otimes \mathrm{id}_C"] \\ C \otimes_R C \arrow[r, "\mathrm{id}_C \otimes \varepsilon"'] & C \end{tikzcd}$$

Example 7.1.2
Let R be a commutative ring. Consider the graded R-module $R[x]$. If $\Delta : R[x] \to R[x] \otimes_R R[x]$ is a graded comultiplication map, then the graded condition forces

$$\Delta(x) = x \otimes 1 + 1 \otimes x$$

Then there are two cases.
- If $\deg(x)$ is even or $2 = 0$ in R, then we have

$$\Delta(x^n) = \sum_{i=0}^{n} \binom{n}{i} x^i \otimes x^{n-i}$$

- Otherwise, we have the two formulas

$$\Delta(x^{2n}) = \sum_{i=0}^{n} \binom{n}{i} x^{2i} \otimes x^{2n-2i} \quad \text{and} \quad \Delta(x^{2n+1}) = \sum_{i=0}^{n} \binom{n}{i} \left(x^{2i+1} \otimes x^{2n-2i} + x^{2i} \otimes x^{2n-2i+1}\right)$$

Definition 7.1.3 (Cocommutative Coalgebra)
Let R be a commutative ring. Let C be a coalgebra over R. We say that R is cocommutative if

$$\tau \circ \Delta = \Delta$$

where $\tau : C \otimes_R C \to C \otimes_R C$ is the swap map.

Definition 7.1.4 (Coalgebra Morphism)
Let R be a commutative ring. Let C, D be coalgebras over R. Let $f : C \to D$ be a $f : C \to D$ be an R-module homomorphism. We say that f is a coalgebra morphism if

$$(f \otimes f) \circ \Delta_C = \Delta_D \circ f \quad \text{and} \quad \varepsilon_D \circ f = \varepsilon_C$$

Definition 7.1.5 (Subcoalgebras)
Let R be a commutative ring. Let C be a coalgebra over R. An R-subcoalgebra is an R-submodule D that is an R-coalgebra in its own right.

Lemma 7.1.6
Let R be a commutative ring. Let C be a coalgebra over R. Let D be an R-submodule D. Then D is an R-subcoalgebra if and only if $\Delta(D) \subseteq D \otimes_R D$.

Definition 7.1.7 (Coideals)
Let R be a commutative ring. Let C be a coalgebra over R. Let I be an R-submodule of C. We say that I is a coideal if the following are true.
- $I \subseteq \ker(\varepsilon)$.
- $\Delta(I) \subseteq I \otimes C + C \otimes I$.

Definition 7.1.8 (Quotient Algebras)

Isomorphism theorems for coalgebras.

7.2 Comodules

Definition 7.2.1 (Comodules)
Let R be a commutative ring. Let C be a coalgebra over R. A left comodule over C is an R-module M together with an R-module homomorphism $\rho : M \to C \otimes_R M$ such that the following diagrams are commutative:

$$\begin{tikzcd} M \arrow[r, "\rho"] \arrow[d, "\rho"'] & C \otimes_R M \arrow[d, "\rho\otimes\mathrm{id}_C"] \\ C \otimes_R M \arrow[r, "\mathrm{id}_M\otimes\Delta"'] & C \otimes_R C \otimes_R M \end{tikzcd} \qquad \begin{tikzcd} M \arrow[r, "\rho"] \arrow[dr, "\mathrm{id}_M"'] & C \otimes_R M \arrow[d, "\varepsilon\otimes\mathrm{id}_M"] \\ & M \end{tikzcd}$$

7.3 Graded Co-Structures

Definition 7.3.1 (Graded Coalgebras)
Let R be a commutative ring. Let (C, Δ, ε) be a coalgebra. We say that R is a graded coalgebra if there exists a decomposition of C into R-modules

$$C = \bigoplus_{n=0}^{\infty} C_n$$

such that the following are true.
- $\Delta(C_n) \subseteq \bigoplus_{i+j=n} C_i \otimes C_j$.
- $\varepsilon(C_n) = 0$ for all $n \geq 1$.

Definition 7.3.2 (Graded Cocommutative Coalgebra)
Let R be a commutative ring. Let C be a graded coalgebra over R. We say that C is graded cocommutative if

$$\Delta = \tau \circ \Delta$$

where $\tau : C \otimes_R C \to C \otimes_R C$ given by $\tau(x \otimes y) = (-1)^{ij} y \otimes x$ for all $x \in C_i$ and $y \in C_j$.

8 Bialgebras and Hopf Algebras

8.1 Bialgebras

Definition 8.1.1 (Bialgebras)
Let R be a commutative ring. An R-bialgebra is an R-module B together with the following data:
- An R-algebra structure (R, μ, η).
- An R-coalgebra structure (R, Δ, ε).

such that the following diagrams are commutative:

$$\begin{tikzcd}
B \otimes_R B \arrow[r, "\mu"] \arrow[d, "\Delta\otimes\Delta"'] & B \arrow[r, "\Delta"] & B \otimes_R B \\
B \otimes_R B \otimes_R B \otimes_R B \arrow[rr, "\mathrm{id}_B\otimes\tau\otimes\mathrm{id}_B"'] & & B \otimes_R B \otimes_R B \otimes_R B \arrow[u, "\mu\otimes\mu"']
\end{tikzcd}
\qquad
\begin{tikzcd}
R \arrow[r, "\eta"] \arrow[dr, "\mathrm{id}_R"'] & B \arrow[d, "\varepsilon"] \\
& R
\end{tikzcd}$$

$$\begin{tikzcd}
B \otimes_R B \arrow[r, "\mu"] \arrow[dr, "\varepsilon\otimes\varepsilon"'] & B \arrow[d, "\varepsilon"] \\
& R
\end{tikzcd}
\qquad
\begin{tikzcd}
R \arrow[r, "\eta"] \arrow[dr, "\eta\otimes\eta"'] & B \arrow[d, "\Delta"] \\
& B \otimes_R B
\end{tikzcd}$$

8.2 Hopf Algebra

Definition 8.2.1 (Hopf Algebras)
Let R be a commutative ring. Let $(H, m, u, \Delta, \varepsilon)$ be an R-bialgebra. Let $S : H \to H$ be an R-module homomorphism. We say that $(H, m, u, \Delta, \varepsilon, S)$ is a Hopf algebra if the following diagram commutes:

$$\begin{tikzcd}
& H \otimes_R H \arrow[rr, "S\otimes\mathrm{id}_H"] & & H \otimes_R H \arrow[dr, "m"] & \\
H \arrow[ur, "\Delta"] \arrow[dr, "\Delta"'] \arrow[rr, "\varepsilon"] & & R \arrow[rr, "u"] & & H \\
& H \otimes_R H \arrow[rr, "\mathrm{id}_H\otimes S"'] & & H \otimes_R H \arrow[ur, "m"'] &
\end{tikzcd}$$

Example 8.2.2
Let G be a group. Then $k[G]$ is a cocommutative Hopf algebra with the following data.
- The comultiplication map $\Delta(g) = g \otimes g$ for all $g \in G$.
- The counit map $\varepsilon(g) = 1$ for all $g \in G$.
- The inversion map $S(g) = g^{-1}$ for all $g \in G$.

Proposition 8.2.3
Let R be a commutative ring. Let H be a Hopf algebra. If H is a projective and finitely generated R-modue, then $\mathrm{Hom}_R(H, R)$ is a Hopf algebra. Moreover, the following are true.
- $\mathrm{Hom}_R(H, R)$ is commutative if and only if H is cocommutative.
- $\mathrm{Hom}_R(H, R)$ is cocommutative if and only if H is commutative.

Hopf subalgebra Hopf ideal Quotient by hopf ideal is a hopf algebra. First isomorphism theorem of Hopf algebra

Tensor product of Hopf algebra is a Hopf algebra

Definition 8.2.4 (Primitive Elements)
Let R be a commutative ring. Let H be a Hopf algebra. Define the primitive elements of H to be

$$P(H) = \{x \in H \mid \Delta(x) = x \otimes 1 + 1 \otimes x\}$$

Definition 8.2.5 (Group-Like Elements)
Let R be a commutative ring. Let H be a Hopf algebra. Define the group-like elements of H to be

$$G(H) = \{x \in H \mid \Delta(x) = x \otimes x\}$$

Dual of Hopf algebra is Hopf algebra.

Dual of the polynomial ring is the divided power algebra (see Hatcher)

8.3 Graded Hopf Algebras

Definition 8.3.1 (Graded Hopf Algebras)
Let R be a commutative ring. Let H be a Hopf algebra over R. We say that H is graded if there exists a decomposition of H into R-modules

$$H = \bigoplus_{k=0}^{\infty} H_k$$

such that the following are true.
- H is a graded algebra with the given decomposition.
- H is a graded coalgebra with the given decomposition.
- $S(H_n) \subseteq H_n$.

8.4 Hopf Algebra over a Field

Theorem 8.4.1 (Hopf 1941)
Let k be a field of characteristic 0. Let H be a graded commutative and graded cocommutative Hopf algebra over k such that each piece H_n is finite dimensional. Then there exists a graded vector space V over k such that

$$H \cong S^{\text{gr}}(V)$$

as graded Hopf algebras.

Theorem 8.4.2 (Milnor-Moore Theorem)
Let k be a field of characteristic 0. Let H be a connected, graded cocommutative Hopf algebra over k. Then the following are true.
- $P(H)$ is a graded Lie algebra.
- There is an isomorphism of Hopf algebras

$$U(P(H)) \cong H$$

  given by the canonical map.

9 The Dual of Algebraic Structures

9.1 Comonoid Objects

9.2 Cogroup Objects

9.3 Coring Objects

9.4 Comodule Objects

9.5 Coalgebra Objects

9.6 Comodules

Let R be a ring. Let M be a left R-module. Since

$$_R\mathbf{Mod} = {_R\mathbf{Mod}}(\mathbf{Set}) = {_R\mathbf{Act}}(\mathbf{Ab})$$

can be thought of as a module object, we can inverse all arrows in the definition of a module to obtain the notion of a comodule.

Careful: bialgebra later means algebra + coalgebra, but biomodules mean left module + right module and does not mean module + comodule.

9.7 Coalgebra

Part IV

Differential Structures

10 Differential Graded Algebras

10.1 Differential Graded Algebras

Definition 10.1.1 (Differential Graded Algebras)
Let R be a commutative ring. A differential graded R-algebra consists of the following data.
- A $\mathbb{Z}$-graded R-algebra A.
- An R-algebra homomorphism of degree -1 (called homological dgas) or of degree 1 (called cohomological dgas).

such that the following are true.
- Differential: $d \circ d = 0$.
- Graded Leibniz: $d(ab) = (da)b + (-1)^{\deg(a)}a(db)$ for a, b homogeneous elements of A.

The differential property means that any (homological) differential graded algebra decomposes into a chain complex whose differential is given by d.

Lemma 10.1.2
Let R be a commutative ring. Let $(A^\bullet, d) \in \mathbf{Ch}^\bullet(R)$ be a cochain complex. Then $A^\bullet$ is a cohomological differential graded algebra if and only if it is a monoid object in $\mathbf{Ch}^\bullet(R)$.

Definition 10.1.3 (Homomorphisms of Differential Graded Algebras)
Let R be a commutative ring. Let A, B be cohomological differential graded algebras. Let $f : A \to B$ be a function. We say that f is homomorphism if the following are true.
- f is a degree 0 morphism of graded algebras.
- f is a cochain map.

Lemma 10.1.4
Let R be a commutative ring. Let (A, d) be a cohomological differential graded algebra. Then $H^*(A, d) = \bigoplus_{n\in\mathbb{Z}} H^n(A, d)$ is a graded ring with multiplication defined by

$$[u] \cdot [v] = [uv]$$

for $u \in H^n(A, d)$ and $v \in H^m(A, d)$.

Proof
Notice that by the graded Leibniz rule, we have $d(uv) = (du)v + (-1)^n ud(v) = 0$ since u and v are cocycles. Hence uv is also a cocycle. ∎

Products of DGA, tensor products, hom set

Definition 10.1.5 (Commutative Differential Graded Algebras)
Let R be a commutative ring. Let A be a differential graded algebra. We say that A is commutative (cdga) if A is graded commutative.

Lemma 10.1.6
Let R be a commutative ring. Let M be an R-module. Let $\varphi : M \to R$ be an R-module homomrphism. Then $K_\bullet(\varphi)$ is a commutative differential graded algebra.

Proposition 10.1.7
Let k be a field. Let A, B be cohomological differential graded algebras. Then the following are true.
- There is a natural isomorphism $H^*(A) \otimes_k H^*(B) \cong H^*(M \otimes_k N)$.
- There is a natural isomorphism $H^*(\text{Hom}_k(A, B)) \cong \text{Hom}_k(H^*(A), H^*(B))$

Definition 10.1.8 (Simply Connected DGAs)
Let R be a commutative ring. Let A be a differential graded algebra. We say that A is simply connected if the following are true.
- $A^0 = R$.
- $A^1 = 0$.

10.2 Minimal DGAs

Definition 10.2.1 (Minimal Differential Graded Algebras)
Let R be a commutative ring. Let A be a differential graded R-algebra. We say that A is minimal if the following are true.
- $d(A_+) \subseteq A_+ \cdot A_+$.
- There is a graded R-module M such that $A \cong T(M)$.

10.3 Massey Products

Definition 10.3.1 (Massey Triple Products)
Let R be a commutative ring. Let A be a differential graded algebra. Let $u \in A^p$, $v \in A^q$ and $w \in A^r$. Suppose that $[u] \cdot [v] = [v] \cdot [w] = 0$. Define the Massey triple product of the three elements to be

$$\langle [u], [v], [w] \rangle = \{ [\overline{s} \cdot w + \overline{u} \cdot t] \in H^{p+q+r-1}(A) \mid s \in A^{p+q-1}, t \in A^{q+r-1} \text{ and } ds = \overline{u} \cdot v, dt = \overline{v} \cdot w \}$$

where $\overline{a} = (-1)^{1+\deg(a)} a$ for any homogeneous element $a \in A$.

Lemma 10.3.2
Let R be a commutative ring. Let A be a differential graded algebra. Let $u \in A^p$, $v \in A^q$ and $w \in A^r$. Suppose that $[u] \cdot [v] = [v] \cdot [w] = 0$. Then we have

$$\langle [u], [v], [w] \rangle = \frac{H^{p+q+r-1}(A)}{[u] \cdot H^{q+r-1}(A) + H^{p+q-1}(A) \cdot [w]}$$

Proposition 10.3.3
Let $p, q, r \in \mathbb{N}$. Let $Y = S^p \vee S^q \vee S^r$. Let $\alpha \in \pi_p(Y)$, $\beta \in \pi_q(Y)$ and $\gamma \in \pi_r(Y)$ be generators. Let $X = Y \cup_{[\alpha,[\beta,\gamma]]} D^{p+q+r-1}$. Let $u, v, w, z \in H^*(X)$ be generators of degree $p, q, r, p+q+r-1$ respectively. Then the following are true.
- $\langle u, v, w \rangle = (-1)^p z$.
- $\langle v, w, u \rangle = (-1)^{pq+pr+p} z$.
- $\langle w, u, v \rangle = 0$.

Part V

Kosuzl Duality

11 Koszul Algebras

11.1 Koszul Algebras

Definition 11.1.1 (Koszul Algebras)
Let k be a field. Let R be a standard graded algebra over k. We say that R is a Koszul algebra if

$$\dim_k(\text{Tor}_i^R(M,k)) = 0$$

for all $i \neq j$.

Let k be a field. Let R be a Koszul algebra over R. Then R admits a minimal graded free resolution $F_\bullet$ and by the constraint of being Koszul, we must have $F_i = R(-i)^{k_i}$ for some $k_i \in \mathbb{N}$.

Example 11.1.2
Let k be a field. Then $k[x_1, \dots, x_n]$ is a Koszul algebra.

Proof
The Koszul complex $K_\bullet(x_1, \dots, x_n, k)$ gives a minimal graded free resolution that one can check that the ith position is given by $R(-i)^{\binom{n}{i}}$. ■

Example 11.1.3
Let k be a field. Then $\frac{k[x,y]}{(x,y)}$ is a Koszul algebra.

Proposition 11.1.4
Let k be a field of characteristic 0. Let V be a finite dimensional vector space over k. Then the following are true.
- $S(V)$ is a Koszul algebra.
- $\Lambda(V)$ is a Koszul algebra.

11.2 Linear Resolutions

We generalize the condition that defines Koszul algebras.

Definition 11.2.1 (Linear Resolutions)
Let k be a field. Let R be a standard graded algebra over k. Let M be a finitely generated graded R-module. Let $F_i = \bigoplus_{j\in\mathbb{Z}} R(-j)^{b_{i,j}(M)}$ be the minimal graded free resolution of M. We say that it is linear if there exists $d \in \mathbb{N}$ such that

$$b_{i,j}(M) = 0$$

for all $j \neq i + d$.

Lemma 11.2.2
Let k be a field. Let R be a standard graded algebra over k. Let M be a finitely generated graded R-module. If M admits a linear resolution $F_i = \bigoplus_{j\in\mathbb{Z}} R(-j)^{b_{i,j}(M)}$ such that $b_{i,j}(M) = 0$ for all $j \neq i + d$, then M is generated at a single degree d.

Definition 11.2.3 (Minimal Resolutions of Type $N_{d,p}$)
Let R be a Notherian $\mathbb{N}$-graded ring such that R_0 is a field. Let M be a finitely generated R-module. Let $F_i = \bigoplus_{j\in\mathbb{Z}} R(-j)^{b_{i,j}(M)}$ be a minimal graded free resolution of M. We say that M is of type $N_{d,p}$ if the following are true.
- M is generated in degree d.
- $b_{i,j}(M) = 0$ for all $j \neq i + d$ when $i < p$.

Example 11.2.4
Let R be a Notherian $\mathbb{N}$-graded ring such that R_0 is a field. Let M be a finitely generated R-module. Let $F_i = \bigoplus_{j\in\mathbb{Z}} R(-j)^{b_{i,j}(M)}$ be a minimal graded free resolution of M.
- M is of type $N_{d,1}$ if it is generated in degree d.
- M is of type $N_{d,2}$ if it is generated in degree d and it is linearly presented.
- M has a linear resolution if and only if M is of type $N_{d,p}$ for all p.

Definition 11.2.5 (Almost Linear Resolutions)
Let R be a Notherian $\mathbb{N}$-graded ring such that R_0 is a field. Let M be a finitely generated R-module. We say that M has an almost linear resolution if M is of type $N_{d,\text{pd}_R(M)}$.

Definition 11.2.6 (Componentwise Linear)
Let R be a Notherian $\mathbb{N}$-graded ring such that R_0 is a field. Let M be a finitely generated R-module.
- We say that M is componentwise linear if $\langle M_d\rangle$ has a liner resolution for all d.
- We say that M is componentwise of type $N_{d,p}$ if $\langle M_d\rangle$ satisfies $N_{d,p}$ for all d.

11.3 Koszul Modules

Definition 11.3.1 (Koszul Modules)
Let k be a field. Let R be a standard graded algebra over k. Let M be a finitely generated graded R-module. We say that M is a Koszul module if M admits a componentwise linear resolution.

The notion of Koszul modules generalize Koszul algebras. Let k be a field. Let R be a standard graded algebra over k. Then clearly R is a Koszul algebra if and only if k is a Kosuzl R-module.

There is an equivalent way to characterize Koszul modules.

Definition 11.3.2 (m-Adic Filtration of the Minimal Graded Free Resolution)
Let k be a field. Let R be a standard graded algebra over k. Let M be a finitely generated graded R-module. Let $(F_\bullet, d)$ be the minimal graded free resolution of M. Define the m-adic filtration of $F_\bullet$ to be the subcomplexes given by

$$(F^jF)_i = m^{j-i}F_i$$

for all $i\in\mathbb{Z}$ and $j\in\mathbb{N}$.

We can then form the associated graded free complex.

Definition 11.3.3 (The Linear Part)
Let k be a field. Let R be a standard graded algebra over k. Let M be a finitely generated graded R-module. Let $(F_\bullet, d)$ be the minimal graded free resolution of M. Define the linear part of $F_\bullet$ to be the associated graded complex

$$\text{lin}(F) = \text{gr}(F^\bullet F)$$

Let M be a graded R-module. We denote

$$M_{\langle\leq n\rangle} = \left\langle \bigoplus_{i=0}^n M_i \right\rangle$$

We can describe the linear part explicitly using the following construction:

- Define the d-linear strand of F to be the chain complex

$$(F_{\langle d\rangle})_n = \frac{(F_n)_{\langle\leq d+n\rangle}}{(F_n)_{\langle\leq d-1+n\rangle}} = \frac{\langle\bigoplus_{i=0}^{d+n}(F_n)_i\rangle}{\langle\bigoplus_{i=0}^{d-1+n}(F_n)_i\rangle}$$

- Then we have
$$\text{lin}(F) = \bigoplus_{d\in\mathbb{Z}} F_{\langle d\rangle}$$

Intuitively, $\text{lin}(F)$ is simply the complex with $\text{lin}(F)_n = F_n$ and with differential obtained by deleting all nonlinear entries from homogeneous matrices representing the differential of F.

Theorem 11.3.4 (Romer 2001)
Let k be a field. Let R be a standard graded algebra over k. Let M be a finitely generated graded R-module. Then M is a Koszul module if and only if M admits a minimal graded free resolution $F_\bullet$ such that $\text{lin}(F_\bullet)$ is acyclic.

11.4 The Local Case

Definition 11.4.1 (m-Adic Filtration of the Minimal Free Resolution)
Let (R,m) be a Noetherian local ring. Let M be a finitely generated R-module. Let $(F_\bullet, d)$ be the minimal free resolution of M. Define the m-adic filtration of $F_\bullet$ to be the subcomplexes given by
$$(F^jF)_i = m^{j-i}F_i$$
for all $i\in\mathbb{Z}$ and $j\in\mathbb{N}$.

We can then form the associated graded free complex.

Definition 11.4.2 (The Linear Part)
Let (R,m) be a Noetherian local ring. Let M be a finitely generated R-module. Let $(F_\bullet, d)$ be the minimal free resolution of M. Define the linear part of $F_\bullet$ to be the associated graded complex
$$\text{lin}^R(F) = \text{gr}(F^\bullet F)$$

Explicitly, we have
$$\text{lin}^R(F)_n = \text{gr}_m(F_n)(-n)$$

Definition 11.4.3 (Koszul Modules)
Let (R,m) be a Noetherian local ring. Let M be a finitely generated R-module. We say that M is a Koszul module if $\text{lin}^R(F)$ is acyclic.

Proposition 11.4.4
Let (R,m) be a Noetherian local ring. Let M be a finitely generated R-module. Then the following are equivalent.
- M is Koszul.
- $\text{lin}^R(F)$ is a minimal graded free resolution of $\text{gr}_m(M)$ over $\text{gr}_m(R)$.
- The $\text{gr}_m(R)$-module $\text{gr}_m(M)$ admits a linear resolution.

Let k be a field. Let R be a standard graded ring over k. Let M be a finitely generated graded R-module. Then we have $\text{gr}_m(M)\cong M$ and $\text{gr}_m(R)\cong R$ so the above proposition generalizes the notion of Koszul modules from graded rings to Noetherian local rings.

Definition 11.4.5 (Koszul Local Rings)
Let (R,m) be a Noetherian local ring. We say that R is Koszul if R/m is a Koszul R-module.

12 Quadratic Algebras

13 Koszul Duality